BRIAN AGWORA OMBATI

Junior Data Scientist

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PROFESSIONAL SUMMARY

Data Scientist with hands-on experience building end-to-end predictive systems, financial risk models, and containerized ML pipelines. Proficient in Python, statistical modelling, and cloud-ready Docker deployments. Proven ability to apply advanced techniques including Extreme Value Theory, time-series feature engineering, and econometric analysis to real-world business problems in fintech, climate science, and media analytics. Actively seeking a junior data science or quantitative analyst role where rigorous modelling meets commercial impact.

TECHNICAL SKILLS

Languages	Python (primary), SQL, R, Bash
ML & Modelling	Scikit-learn, XGBoost, Statsmodels, SciPy, Extreme Value Theory (EVT)
Data Engineering	Pandas, NumPy, time-series feature engineering, spatial imputation, rolling statistics
Visualisation	Matplotlib, Seaborn, Plotly
DevOps / Cloud	Docker (containerisation), Git / GitHub, Linux CLI
Finance / Quant	Credit risk modelling, Block Maxima (EVT), tail-risk simulation, volatility benchmarking
Statistics	Hypothesis testing, regression analysis, game theory, econometrics

TECHNICAL PROJECTS

1.) Full-Scale Meteorological Predictive AI System [GitHub]

- Built an end-to-end weather forecasting pipeline in Python with dedicated modules for automated data loading, cleaning, and statistical feature engineering.
- Engineered time-series lagging, spatial imputation, and rolling-window statistical metrics to handle irregular atmospheric data patterns.
- **Docker**Optimised predictive model performance and containerised the full application inside
- Docker deployment enables cross-platform cloud portability — a production-grade workflow beyond typical notebook-based data science projects.

2.) Loan Default Prediction & Risk Architecture [GitHub]

- Developed a credit-scoring model using supervised ML algorithms (Logistic Regression, Gradient Boosting) to assess borrower default probability on consumer financial data.
- Integrated Extreme Value Theory (EVT) via the Block Maxima approach to statistically model tail-risk and simulate catastrophic loss thresholds.
- Benchmarked extreme financial asset volatility by comparing historical tail behaviour of the S&P 500 directly against Kenyan Sovereign Bonds — quantifying cross-market risk divergence.

3.) Dynamic Earth Carrying Capacity Simulation Model [GitHub]

- Designed a multi-parameter mathematical simulation using applied statistical modelling to estimate global resource limits under varying demographic and energy scenarios.
- Consolidated variables including arable land distribution, water scarcity vectors, demographic trajectories, and energy outputs into an interconnected systemic model.

4.) Competitive Rivalry Modelling in the Entertainment Ecosystem [GitHub]

- Structured an econometric analysis exploring market competition effects on consumer retention and corporate revenue streams in the media industry.
- Applied statistical hypothesis testing and game-theory logic to isolate variables driving user churn when competing platforms launch rival content.

EDUCATION

BSc Applied Statistics| **Kisii University** | Kisii, Kenya. *40200*

- Relevant coursework: Statistics, Econometrics, Machine Learning, Financial Mathematics, Database Systems

WORK EXPERIENCE

Data Analyst / Intern (or most recent role) | **Gusii Mwalimu Sacco Lmt** | Location
April 2025 – August 2025

- Increased queueing efficiency by 15%
- Predicted loan defaulters and Reduced the risk by 7%
- Excellent customer service
- Supermarket cashier.

CERTIFICATIONS & CONTINUOUS LEARNING

- Python Programming
- Advanced Excel/SPSS
- Power BI
- Self-directed study: Extreme Value Theory, quantitative risk modelling, Docker & cloud deployment best practices

ADDITIONAL

- Languages: English (professional), Kiswahili, Kegusii
- Interests: Quantitative finance, open-source data science, environmental modelling, competitive strategy
- Available for: Full-time roles, internships, contract / freelance data projects